



Politecnico di Torino

Electromagnetic and Thermo-Hydraulic Design of a Double-Pancake HTS CICC Solenoid for 15 T and 50 kA

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Chapter 1

Introduction

1.1 Project objectives and design specifications

The aim of this project is to design a high-field HTS solenoid, wound as stacked double pancakes, using REBCO coated conductors with CICC cooling layout. The focus of the project is on the double pancake located on the equatorial plane of the solenoid. The feasibility of this design will be demonstrated both by an electro-magnetic and by a thermo-hydraulic stand point.

The headline imposes two main electromagnetic requirements for the solenoid. The first one is for the magnetic field experienced by the conductor, also including the relevant self-field contribution, to reach approximately 15 T . The second one, is a nominal current of 50 kA , to be reached with a constant ramp rate of 0.2 kA/s , resulting in a ramp time of 250 s .

Geometric constraints are also imposed: the winding has an inner diameter $D_{in} = 1\text{ m}$, while the total height of the solenoid must not exceed 1.5 m .

The conductor is cooled using helium, whose operating window is constrained by inlet temperature, pressure and mass flow:

$$4.5\text{ K} \leq T_{in} \leq 50\text{ K} \quad (1.1)$$

$$5\text{ bar} \leq p \leq 20\text{ bar} \quad (1.2)$$

$$\dot{m}_{max} = 15\text{ g/s} \quad (1.3)$$

The main thermal requirement is a minimum temperature margin larger than 5 K . This margin is defined as the difference between the current sharing temperature and the actual temperature of the conductor in all spatial points and throughout the whole transient:

$$T_{cs}(s) - T_{sc}(s, t) \geq 5\text{ K} \quad (1.4)$$

The architecture of the cable uses REBCO tapes arranged in stacks and fitted inside a stacked-slotted CICC, with copper used as an electric stabilizer, an external stainless-steel jacket for structural purposes and a central channel for the supercritical helium to flow into.

Table 1.1: Main project specifications.

Quantity	Value	Origin
Rated current	50 kA	Project brief
Current ramp rate	200 A/s	Project brief
Ramp duration	250 s	Derived
Peak field on conductor	15 T	Project brief
Required thermal margin	5 K	Project brief
Winding inner diameter	1.0 m	Project brief
Maximum solenoid height	1.5 m	Project brief
Maximum total He mass flow	15 g/s	Project brief
He inlet-temperature range	4.5 K to 50 K	Project brief
He pressure range	5 bar to 20 bar	Project brief

1.2 Overall design workflow

The starting point of our workflow was the acquisition and validation of data containing the superconducting properties of the REBCO tapes. These data are obtained through different sources. We referred to the data book for the HERMES tape provided by Faraday Factory to obtain the critical current's dependence on temperature and magnetic field. We then directly contacted the company, which provided to us a more detailed dataset, including angular measurements, from which we obtained the power-law exponent. The experimental measurements carried out during the laboratory activities were instead used to validate the analysis procedure of the I-V curves and the expected qualitative dependence between I_c and n .

Afterwards, the global geometry of the solenoid was addressed, starting from the imposed constraints. The code builds the solenoid as an axial stack of double pancakes, taking into account the prescribed inner diameter, the maximum winding height, the diameter of the cable and the targets for current and magnetic field. The code then selects the configuration that comes closer to the 15 T requirement.

Once the global geometry is fixed, the focus of our analysis shifts to the central DP. At every position along this DP, the code determines the local magnetic field intensity and orientation, dividing it into a parallel and perpendicular component relative to the tape surface. This information, together with an assumed operating temperature, is used to evaluate the local critical current, which in turn gives a spatial profile of the current sharing temperature $T_{cs}(s)$. A first, electromagnetic temperature margin can then be computed, although this is not yet the final thermal margin.

The next step is an analysis of how the current distributes itself between the different strands. The strands are coupled together through inductive interactions and exhibit a non-linear electric behavior which is described by a set of differential-algebraic equations. This model follows the current distribution during the current ramp and demonstrates that current equipartition emerges naturally from the geometry and the electrical equivalence of the different strands.

Once the electromagnetic configuration is established, the code constructs the cross section of the CICC cable, determining the actual positions of the tape stacks, as well as the dimensions of the helium channel and the areas of the different materials which are used in the thermal model. The magnetic field history during the current ramp is then used

to evaluate the AC losses, which are divided into an hysteresis and a coupling term. Lastly, the code uses the information derived so far to evaluate the cooling needs of the cable. This is done by solving a Gandalf-type thermo-hydraulic model, which determines the minimum mass flow to satisfy the thermal margin. The design that we obtained has been checked iteratively for consistency between the different parts.

Chapter 2

Input Data, Tape Characterization, and Experimental Activity

2.1 HERMES tape and supplier data

The reference conductor that we used for our design is the HERMES REBCO coated conductor produced by Faraday Factory. The tape has a width of 4 *mm* and is composed by several layers made out of different materials. The composition as well as the width of each layer are detailed in *Table 2.1*.

Table 2.1: Nominal HERMES tape architecture used in the design model.

Layer	Nominal thickness
Cu, top	5.0 μm
Ag, top	2.0 μm
REBCO	2.5 μm
Buffer stack	0.1 μm
Hastelloy C-276 substrate	40 μm
Ag, back	1.0 μm
Cu, back	5.0 μm
Total nominal thickness	55.6 μm
Tape width	4.0 mm
Average I_c at 77 K, self-field	100 A to 200 A

For this tape, the supplier gives an average self-field critical current at 77 *K* in the range of 100 – 200 *A*. For our project, we decided to adopt the lower end of this range as a conservative choice:

$$I_{c,ref} = I_c(77 \text{ K}, sf) = 100 \text{ A} \quad (2.1)$$

This reference value is then used to evaluate the critical current at different temperatures and magnetic fields, using the values from the lift factor table provided by the manufacturer in the HERMES Data Book. Lift factors are defined as:

$$LF(T, B) = \frac{I_c(T, B)}{I_{c,ref}} \quad (2.2)$$

Meaning that the design critical current can simply be constructed as:

$$I_c(T, B) = I_{c,ref} LF(T, B) \quad (2.3)$$

This table covers a wide range of field values, from self field up to 20 T , with a temperature range that goes from 4.2 K to 77 K . This is why we chose it as our primary database for the critical current evaluation. The table is also conservative with respect to field orientation, as the displayed values are the minimum ones over all measured field angles. The critical current is therefore made independent from the angular orientation.

Table 2.2: Role and coverage of the superconducting-property datasets.

Source	Field range	Temperature range	Statistical basis	Use in this project
HERMES Data Book	0 T to 20 T	4.2 K to 77.3 K	Supplier statistical database	Design basis for $I_c(T, B)$ and lift factors
Faraday dataset	0 T to 8 T	15 K to 77.5 K	One representative tape sample, angular scans	$n(T, B)$ and cross-validation of lift factors
Group-9 laboratory test	field not quantified	~ 76.9 K	One laser-cut bridge sample	Independent methodological check only

The second dataset provided by Faraday only covers field values up to 8 T and is therefore not suitable for the critical current evaluation. We instead used it to evaluate the power law exponent for the nonlinear electrical model of the conductor. The exponent is provided as a function of temperature and magnetic field. This dataset was also useful to cross-check the values from the Data Book in the region where the two datasets overlap.

Table 2.3: Key electrical-property conventions used throughout the model.

Quantity	Value/model	Note
Reference $I_c(77$ K, s.f.)	100 A	Conservative adopted reference
Electric-field criterion E_0	1×10^{-4} V/m	1 μ V/cm, supplier-confirmed
n model	$n(T, B)$ table	Faraday detailed dataset
n measured domain	15 K to 77.5 K, 0 T to 8 T	Clamped at domain edge in evaluator
Lift-factor angular treatment	$\min_{\theta} LF(T, B, \theta)$	Conservative magnet-design basis

The supplier indicates that the properties of the conductors can vary from sample to sample by up to 15 – 20 %. This variability is taken into account in the evaluation of the current-sharing temperature margin.

All of the quantities used in the model are listed in the file `dataAcquisition.m`, together with their origin (headline specifications, supplier data, literature values, laboratory measurements). All of the assumed quantities, like the 100 A value for the reference critical current, are also explicitly marked.

2.2 Experimental activity: I–V characterization of a coated conductor

The superconducting sample used during the laboratory sessions is also a REBCO coated conductor. However, its architecture is not identical to the one of the HERMES project tape, meaning that a direct numerical comparison must be handled with caution.

The voltage measurement across the sample was carried out with the four probe method: two outer probes connected the sample to a power source, while two inner probes were connected to a voltmeter to measure the voltage drop.

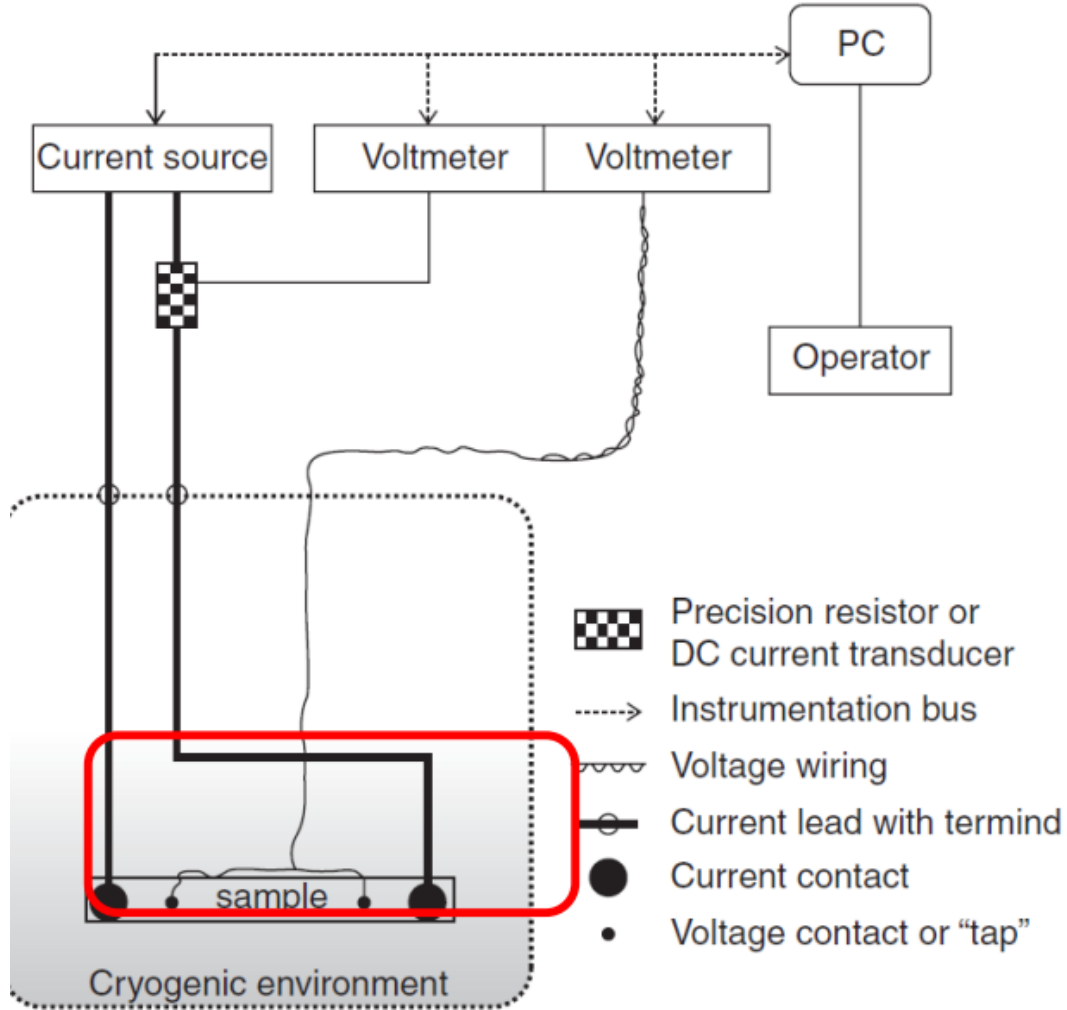


Figure 2.1: Four-probe measurement scheme used for the I–V tests.

Although the tape measured $\approx 4 \text{ mm}$ in width, a narrow bridge of only $\approx 0.3 \text{ mm}$ was patterned using laser cutting. This was done to allow the sample to reach superconducting state with the limited current available in the laboratory.

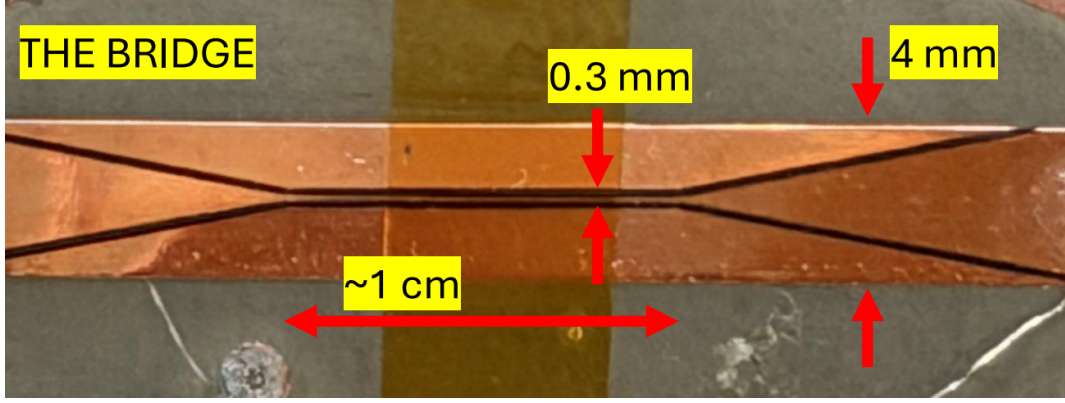


Figure 2.2: Laboratory coated-conductor sample and laser-cut bridge geometry used for transport characterization.

Before the I-V measurements, a cooling test was performed by gradually decreasing the sample's temperature using a liquid nitrogen bath. A superconducting transition was observed at $\approx 86\text{ K}$, quite close to the expected critical temperature of 90 K for REBCO tapes. The standard criterion to identify the transition is:

$$E_c = 1\mu V/cm \quad (2.4)$$

Which converted into a voltage criterion gives:

$$V_c = E_c L_{tap} \quad (2.5)$$

Where L_{tap} is the distance between the two voltage taps in the four-probe setup. We approximated this distance as the length of the narrow bridge, which is 1 cm . This way we obtained critical current values of $I_c \approx 1.94\text{ A}$ in self-field and $I_c \approx 1.07\text{ A}$ with an applied field. However, due to the uncertainty of the tapping distance, these values do not represent certified properties of the tape.

Once the sample's temperature stabilized at $\approx 77\text{ K}$, the first I-V measurement with no external field began. The current was increased in intervals and held at each value for a long enough time to measure the steady voltage after the initial transient caused by the measurement electronics. Even at low currents a small and approximately constant background voltage was recorded. In the self-field measurement this was $V_{bg} \approx 0.24\text{ }\mu V$. The measured voltage was therefore corrected before fitting the superconducting power law:

$$V_{corr} = V_{meas} - V_{bg} \quad (2.6)$$

Once the background is removed, the transition is analyzed using the following power law:

$$V = V_c \left(\frac{I}{I_c} \right)^n \quad (2.7)$$

Taking the logarithm of both terms we obtain:

$$\log V = n \log I + (\log V_c - n \log I_c) \quad (2.8)$$

In a log-log representation, the coefficient represents the slope of the curve;

$$n = \frac{d(\log V)}{d(\log I)} \quad (2.9)$$

The analysis shows that the measured curve is not well represented by a single power law over the whole superconducting transition. Instead, two fitting regions are considered, one at low voltage and one at high voltage. For the self-field tests we obtained:

$$n_1 \approx 17.5 , \quad n_2 \approx 8.1 \quad (2.10)$$

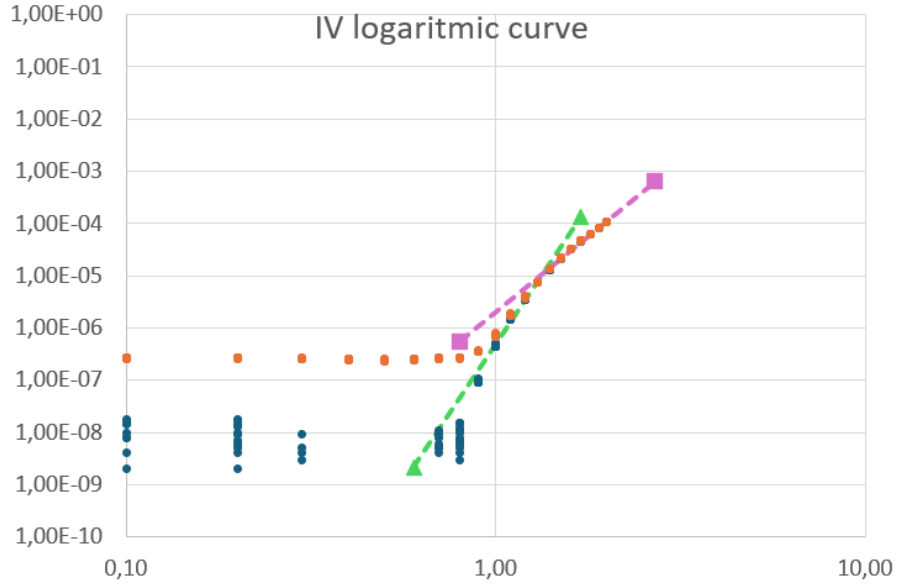


Figure 2.3: Logarithmic current–voltage characteristic measured on the REBCO bridge sample with the permanent magnets applied. The corrected transition region is fitted by two linear regressions in log–log coordinates, giving $n_1 = 10.61$ and $n_2 = 5.81$. The measurement is used only as a qualitative indication of the degradation of superconducting performance in applied magnetic field, since the field magnitude and orientation were not accurately characterized.

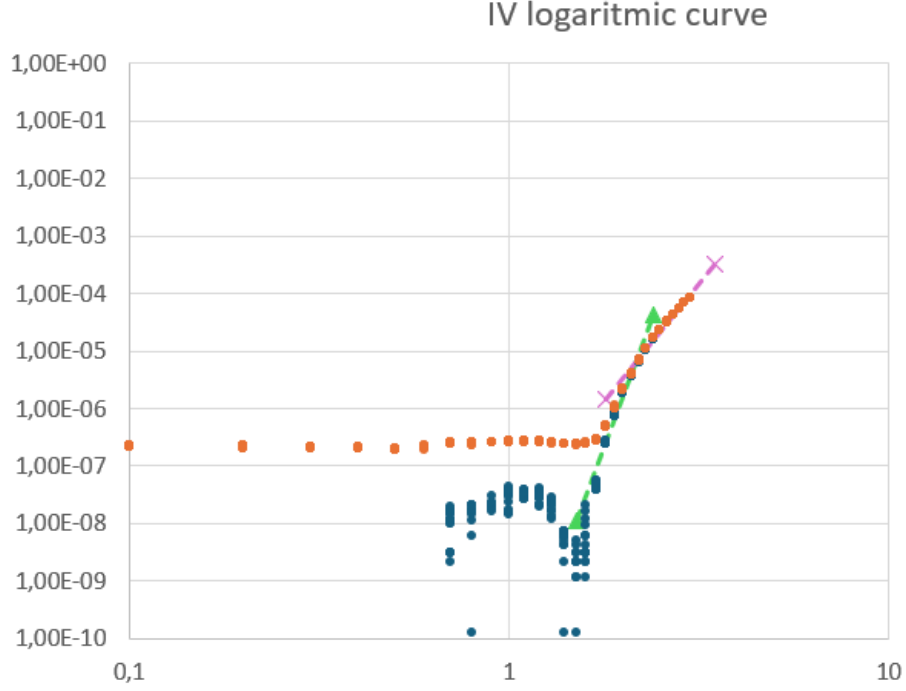


Figure 2.4: Logarithmic current–voltage characteristic measured on the REBCO bridge sample in self-field. The corrected transition region is represented by two linear regressions in log–log coordinates, yielding $n_1 = 17.50$ and $n_2 = 8.12$. The approximately linear portions of the curve confirm the applicability of the power-law representation $V \propto I^n$ over the fitted current ranges.

A second test was also performed by applying an external field to the sample using a permanent magnet. As expected, in this test the recorded critical current was lower than in the self-field test. The fitted values for the coefficients obtained in this test are:

$$n_1 \approx 10.6, n_2 \approx 5.8 \quad (2.11)$$

Thus what we observed is that the application of an external magnetic field degrades the current-carrying capability of the sample as well as the sharpness of its transition from superconducting to normal state.

Table 2.4: Main results of the laboratory I–V analysis.

Quantity	Self-field	Permanent magnet applied
Mean test temperature	76.92 K	76.92 K
Voltage background	0.237 μ V	0.258 μ V
Selected breakpoint current	2.1 A	1.3 A
n_1	17.50	10.61
n_2	8.12	5.81
Criterion current for assumed 1 cm tap spacing	1.94 A	1.07 A

The criterion-current row is illustrative only because the voltage-tap separation was not reliably recorded. These values are not used as design inputs.

From these experimental results, we can observe that the critical current of the sample, even accounting for the reduced width at the narrow bridge, is significantly lower than

the 100 A reference value that we chose for our design. This is a sign than the quality of the sample we used is not up to the industrial REBCO tape standard. For this reason, we will only reference the two datasets provided by Faraday for the calculations involving I_c and n .

Chapter 3

Winding Geometry Design

3.1 Geometric constraints and final winding dimensions

The diameter of the CICC cable was set as:

$$D_{CICC} = 25 \text{ mm} \quad (3.1)$$

If we suppose that there is no additional insulation or spacing between the pancakes, we can write the height of a double pancake as:

$$h_{DP} = 2D_{CICC} = 50 \text{ mm} \quad (3.2)$$

Using the maximum allowed solenoid height of 1.5 m this results in 30 double pancakes. However, we had to use an odd number of double pancakes for the central DP to lie exactly on the equatorial plane of the solenoid. We therefore settled on:

$$N_{DP} = 29, H = 1.45 \text{ m} \quad (3.3)$$

Once N_{DP} is set, the main degree of freedom for the geometry is the number of radial turns N_{turn} . Our code ran various configurations with different integer values for N_{turn} and found the one giving a field value closest to the 15 T target. The final configuration has:

$$N_{turn} = 7 \quad (3.4)$$

Resulting in a peak field on the conductor of:

$$B_{pk} \approx 15.1 \text{ T} \quad (3.5)$$

Starting from the internal radius of the solenoid $R_{in} = 0.5 \text{ m}$, the outer radius is evaluated as:

$$R_{out} = R_{in} + N_{turn}D_{CICC} = 0.675 \text{ m} \quad (3.6)$$

3.2 Global solenoid geometry

As previously mentioned, the central DP is centered at $z = 0$, making the complete winding symmetric about the equatorial plane. For the later electromagnetic model there are two distinct geometric representations. The central DP is discretized on a microscopic level, with all 6 strands being represented separately. The other 28 DPs are instead

represented macroscopically as a single strand. The global solenoid geometry is presented in *Figure 3.1*.

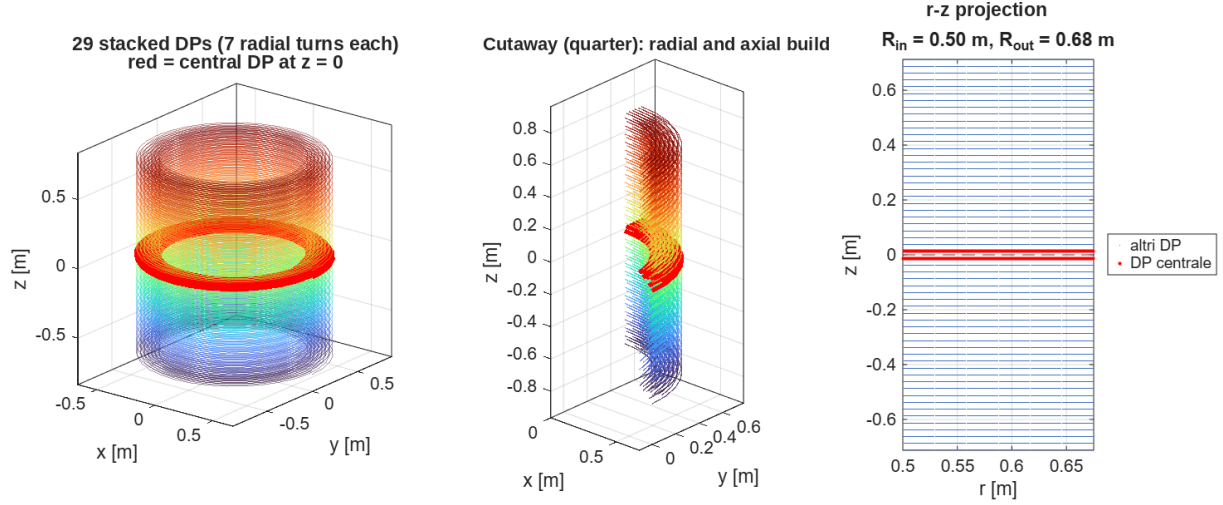


Figure 3.1: Global solenoid geometry: 29 stacked double pancakes, radial/axial build, and r - z projection.

3.3 Double-pancake path and strand twist

In the central DP, the strands have been modeled to have one continuous conductor path through the two pancakes. The current enters at the outer radius of the first pancake, it follows the first spiral to the inner radius where it crosses into the second pancake and then follows the second spiral to the outer radius. The total centerline length of the central DP is:

$$L_{DP} \approx 51.6 \text{ m} \quad (3.7)$$

Each of the six tape slots in the cable, which contains a stack of 80 tapes, is represented by an equivalent strand. The thickness of the stack is not retained, only its position. The six equivalent strand centroids lie around the cable axis at a distance:

$$R_{core} \approx 7.276 \text{ mm} \quad (3.8)$$

For each winding turn, the number of complete twists is set as:

$$N_{twist} = 6 \quad (3.9)$$

The strand angular position is constructed as:

$$\alpha_j(\theta) = \alpha_{j,0} + N_{twist}\theta \quad (3.10)$$

Where $\alpha_{j,0}$ is the initial phase of the six strands, defined as:

$$\alpha_{j,0} = (j - 1) \frac{2\pi}{6} \quad (3.11)$$

During one turn each strand rotates by 12π , meaning six complete turns, and at the end of the turn it returns to the same angular position in the cable cross-section. The geometry is designed to make the six strands equivalent. However, as previously mentioned, the

equipartition of current is demonstrated with the DAE model. A direct consequence of imposing a fixed number of twists per turns is the fact that the local twist pitch l_p varies with the radius:

$$l_p = \frac{2\pi r}{N_{twist}} \quad (3.12)$$

The pitch varies between 0.516 m at the inner radius and 0.707 m at the outer radius. All of the geometrical data is reported in *Table 3.1*.

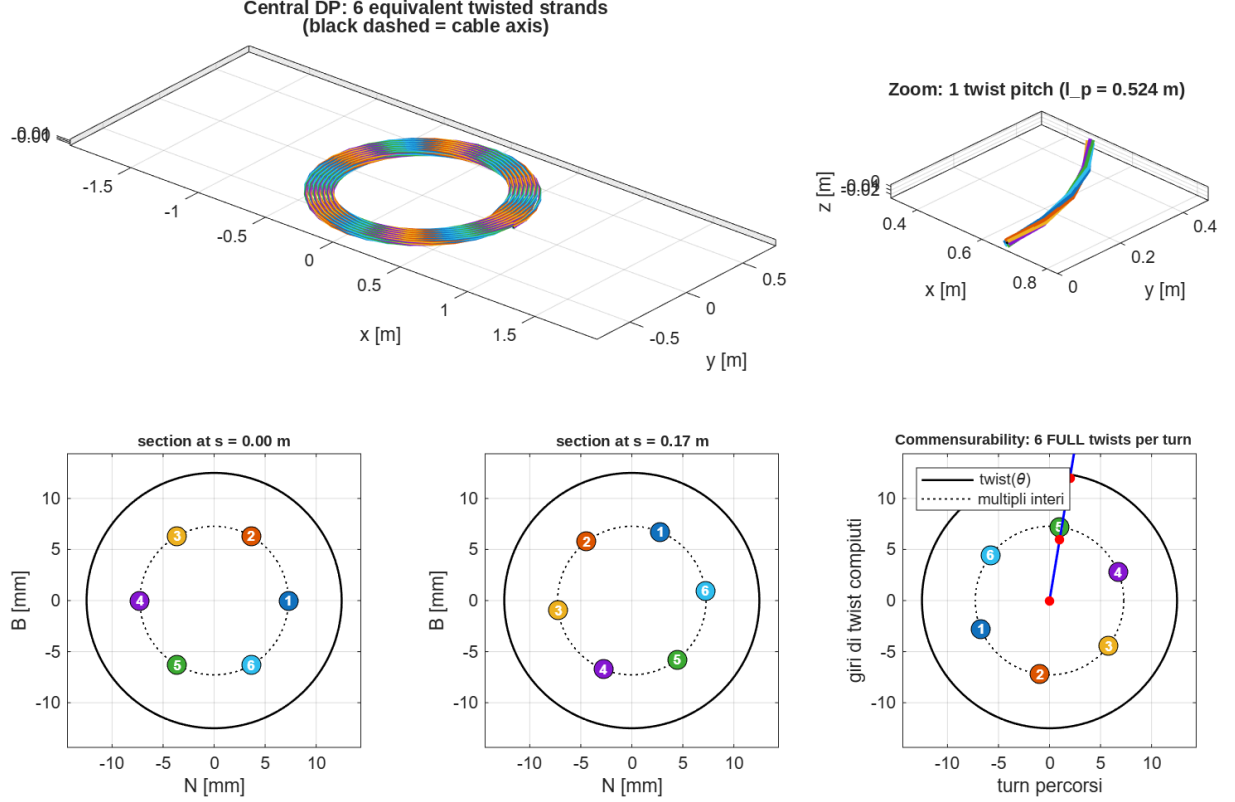


Figure 3.2: Central double pancake with six equivalent twisted strands, one-pitch detail, transverse sections, and twist commensurability check.

Table 3.1: Final winding and conductor-level geometry.

Quantity	Final value
Number of double pancakes	29
Radial turns per pancake	7
Solenoid inner radius	0.500 m
Solenoid outer radius	0.675 m
Total winding height	1.45 m
CICC outer diameter	25 mm
Equivalent CICC slots/strands	6
Tapes per slot	80
Twists per turn	6
Local twist pitch	0.516 m to 0.714 m
Central DP conductor length	51.60 m
Total ampere-turns	2.03×10^8 A-turn

Chapter 4

Magnetic Field, Critical Current, and Current-Sharing Margin

4.1 Magnetic-field calculation

The evaluation of the magnetic field seen by the central DP is carried out at the nominal current $I_{nom} = 50 \text{ kA}$. At every point of one of the six strands the field is decomposed into three contributions:

- External field B_{ext} . This is the contribution generated by the other 28 DPs of the solenoid. It is evaluated using the discretized conductor geometry and the function *segmentMagneticField1d*.
- Mutual field B_{mut} . This is the field produced by the other five strands of the central DP. When evaluating the numerical solution of the segmented Biot-Savart expression on one strand, the strand itself must be excluded from the calculation to avoid a singularity.
- Self-field B_{self} . The self-field of one strand is evaluated analytically by approximating it to a circular conductor:

$$B_{self} = \frac{\mu_0 I_{slot}}{2\pi r_0} \quad (4.1)$$

Where r_0 is the equivalent radius obtained from the cross-sectional area of one tape stack.

The self-field is added conservatively in magnitude, since its exact direction relative to the other contributions is not reconstructed:

$$B_{tot} = |B_{ext} + B_{mut}| + B_{self} \quad (4.2)$$

The peak field value achieved is:

$$B_{pk} = 15.07 \text{ T} \quad (4.3)$$

This value occurs at the innermost radial turn, as is to be expected in solenoid geometry. Because of the twist, the orientation of each tape stack relative to the global magnetic field changes continuously. The model constructs the local normal to the tape $\mathbf{n}_{face}$ and evaluates the two field components:

$$B_{\perp} = |\mathbf{B} \cdot \mathbf{n}_{face}|, \quad B_{\parallel} = \sqrt{|\mathbf{B}|^2 - B_{\perp}^2} \quad (4.4)$$

The corresponding angle is then evaluated as:

$$\theta = \tan^{-1} \left(\frac{B_{\perp}}{B_{\parallel}} \right) \quad (4.5)$$

These components are not required for the critical current evaluation, since the lift factor table only requires field magnitude and temperature. However, the perpendicular component is later used for the evaluation of hysteresis AC losses. It's also important to specify that this decomposition only involves the mutual and external components of the field, while the self contribution is always used in magnitude.

Since the six equivalent strands are geometrically located at the centroid of the tape stacks, the difference in magnetic field from one end of a stack to the other is not considered. Since the inner edge of the stacks is closer to the surrounding strands, it experiences a slightly higher local field. As a conservative correction, the code introduces a coefficient for the mutual field component:

$$C_{mut} = \frac{R_{core}}{r_{near}}, \quad r_{near} = R_{core} - \frac{d_{stack}}{2} \quad (4.6)$$

The field used for I_c and T_{cs} is therefore:

$$B_{near} = |B_{ext}| + C_{mut}|B_{mut}| + B_{self} \quad (4.7)$$

The correction is not needed for the external contribution, since the distance between the strand and the other DPs is much larger than the stack width.

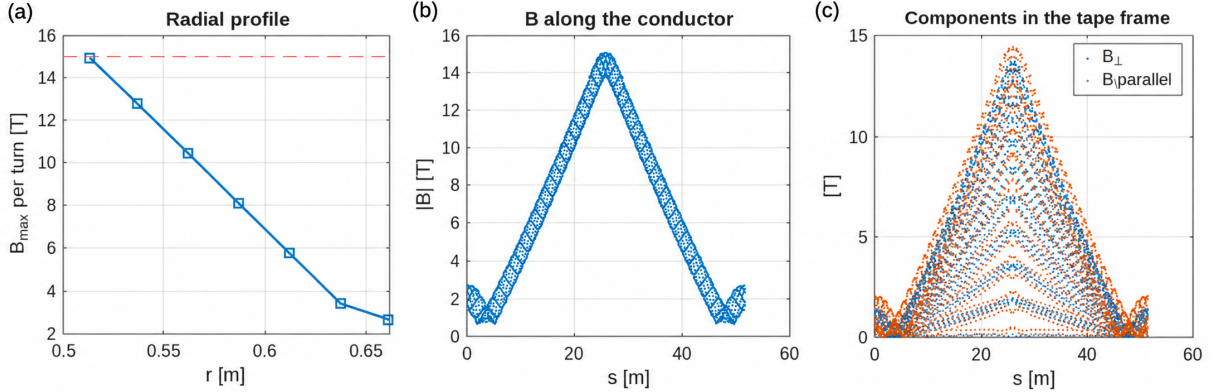


Figure 4.1: Magnetic-field distribution in the central double pancake at the nominal transport current. (a) Maximum magnetic-field magnitude in each radial turn, showing the highest field on the innermost turn and the progressive decrease toward the outer radius; the dashed line marks the 15 T design target. (b) Magnetic-field magnitude along the curvilinear coordinate s of the conductor, highlighting the high-field region close to the inner turns of the double pancake. (c) Decomposition of the local field into the components perpendicular and parallel to the tape face, $B_{\perp}$ and $B_{\parallel}$, accounting for the strand twist along the conductor. These field profiles are subsequently used to evaluate the local critical current and, in the case of $B_{\perp}$, the magnetization-loss contribution.

As can be seen in the figures, the initial and final sections of the DP experience an inversion in the magnetic field trend. This is due to the fact that the external field goes almost to zero and the self-field becomes the most relevant component. However, at the DP edges the self-field is affected by an error because the model does not account for inter-pancake joints and the geometry is not closed.

4.2 Critical current and current-sharing temperature

For this evaluation, the nominal current is assumed to be equally divided between the six equivalent strands. The equipartition is then demonstrated in chapter 5. The reference operating temperature adopted for the electromagnetic calculations is:

$$T_{op} = 16.46 \text{ K} \quad (4.8)$$

This value was obtained after closing the electromagnetic and thermo-hydraulic consistency loop. Using the lift factor table, we obtain a spatial range of critical current values:

$$146.8 \text{ A} \leq I_c(s) \leq 472.7 \text{ A} \quad (4.9)$$

The large spatial variation is a consequence of the variation of the magnetic field along the double pancake. A dimensionless quantity is defined to evaluate how close the current flowing through the tape is to the local critical value:

$$f_I(s) = \frac{I_{tape}}{I_c(s)} \quad (4.10)$$

The maximum is approximately $f_{I,max} = 0.71$, meaning that the most loaded tape operates at 71% of the critical current at T_{op} .

The current-sharing temperature T_{cs} is defined, at fixed field and transport current, as the temperature at which the superconducting tape reaches its critical-current criterion:

$$I_c(T_{cs}, B) = I_{tape} \quad (4.11)$$

Since the magnetic field changes in space, we also obtain a spatial profile $T_{cs}(s)$. For each point along the double pancake, this value is obtained through a bisection method, comparing with the values of I_c provided by the manufacturer. An issue we encountered is the fact that at high fields the supplier data book only provides values for low temperatures. In the final run, only 1352 out of 2694 nodes have a certified value for T_{cs} , while all the others use the highest provided value as a conservative estimate. This can be observed in the plots for T_{cs} , where the inner points of the pancake are displayed as having $T_{cs} = 20 \text{ K}$. This is the highest value provided by the data book for high magnetic fields. The minimum certified current sharing temperature is:

$$T_{cs,min} = 35.98 \text{ K} \quad (4.12)$$

Table 4.1: Main electromagnetic design results from the completed reference run.

Quantity	Value
Peak field on conductor	15.07 T
Operating temperature used in EM/loss model	16.46 K
Current per slot	8333 A
Current per tape	104.2 A
Local tape I_c range at T_{op}	146.8 A to 472.7 A
Maximum I_{tape}/I_c	0.710
Critical certified T_{cs}	35.98 K
Minimum certified current-sharing margin	19.52 K
Certified T_{cs} nodes	1352 / 2694

4.3 Current-sharing margin and lift-factor robustness

The code defines two separate temperature margins. The static current-sharing margin measures the distance between the reference operating temperature in the electromagnetic model and the current sharing temperature:

$$\Delta T_{cs,op}(s) = T_{cs}(s) - T_{op} \quad (4.13)$$

Using the minimum certified value for T_{cs} , we obtain a minimum margin of:

$$\Delta T_{cs,min} = 19.52 \text{ K} \quad (4.14)$$

This, however, is not the final thermal verification required by the project, which is instead evaluated in Chapter 7.

Another aspect to consider is the statistical spread of approximately 15 – 20% in lift factors from sample to sample. This is taken into account by reducing the lift factors by 20% as a conservative measure. The corresponding current-sharing temperature is then used to determine the reference operating temperature through:

$$T_{op} = T_{cs-20\%} - 5 \text{ K} \quad (4.15)$$

This is what ultimately results in the value $T_{op} = 16.46 \text{ K}$.

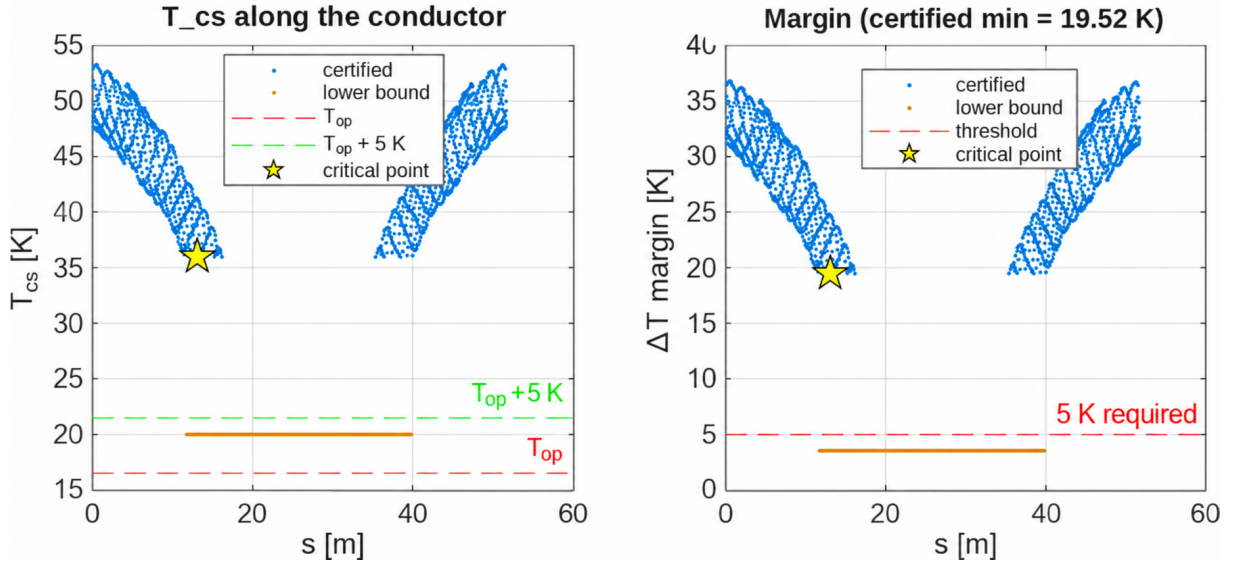


Figure 4.2: Current-sharing temperature and corresponding static temperature margin along the central double pancake. Left: $T_{cs}(s)$ obtained from the condition $I_c(T_{cs}, B) = I_{tape}$. Blue points indicate values for which the crossing is found within the supplier-data temperature range, while orange points represent lower bounds where the actual T_{cs} lies above the available measurements. The reference operating temperature T_{op} and the $T_{op} + 5$ K threshold are also shown. Right: corresponding current-sharing margin $\Delta T = T_{cs} - T_{op}$; the minimum certified value is 19.52 K. Orange values must be interpreted as lower bounds rather than as calculated margins.

Chapter 5

DAE Model and Demonstration of Current Equipartition

5.1 DAE formulation and nonlinear power law

The purpose of the DAE analysis is to show that the current naturally divides itself into 6 equal currents in the tape slots. The model describes the discretized conductor as a set of 2 unknowns: the edge currents i and the nodal electric potentials ϕ . The state vector is therefore:

$$\mathbf{x} = \begin{bmatrix} \mathbf{i} \\ \phi \end{bmatrix}, \quad (5.1)$$

The mesh is composed of 24 segments per winding turn, corresponding to four segments per twist. The full DAE equation is:

$$\begin{bmatrix} \mathbf{L} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} \mathbf{i} \\ \phi \end{bmatrix} + \begin{bmatrix} \mathbf{R}(\mathbf{i}) & \mathbf{A} \\ -\mathbf{A}^T & \mathbf{G} \end{bmatrix} \begin{bmatrix} \mathbf{i} \\ \phi \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{i}_s \end{bmatrix}. \quad (5.2)$$

The inductance matrix $\mathbf{L}$ contains the self and mutual inductive coupling between the conductor segments and is calculated through the function *inductanceMatrix1d*. It controls the inductive current redistribution during the ramp. The incidence matrix $\mathbf{A}$ describes the topology of the electrical network. It connects the edge currents with the nodal voltages, enforcing Kirchoff's current law. The mass matrix is defined as:

This matrix is singular because the currents possess a differential term, while the nodal potentials can be instantly determined by the algebraic Kirchhoff constraints. This is what makes it a differential-algebraic system of equations.

The law describing the superconducting transition is:

$$E = E_0 \left(\frac{I}{I_c} \right)^n \quad (5.3)$$

With $E_0 = 10^{-4} \text{ V/m} = 1 \text{ } \mu\text{V/cm}$ For an electrical segment of length l we have:

$$V = El = RI \quad (5.4)$$

Therefore, the equivalent nonlinear resistance will be:

$$R(I) = \frac{E_0 l}{I_c} \left(\frac{I}{I_c} \right)^{n-1} \quad (5.5)$$

Since R depends in the current, the problem must be solved iteratively. The exponent n determines how sharply the conductor transitions from superconducting to normal state. Its value is evaluated locally at each edge position, keeping the fixed reference temperature $T_{op} = 16.46 \text{ K}$. The main limitation of the dataset that we used to evaluate n , is the fact that it only provides data up to $B = 8 \text{ T}$. Above the tabulated range, the code fixes n at the boundary value. This is a limitation of our model, however its effect on the DAE solution is negligible. This is because the conductor operates at nominal conditions, so sufficiently below I_c , making the resistive voltage much smaller than the inductive contribution.

5.2 Boundary conditions and time integration

At the injection face, a Dirichlet boundary condition is imposed, with all six strands having the same potential at this terminal. The potential at the extraction face is also imposed to be equal for all strands, but its value is not known beforehand, so we have a floating equipotential boundary. The total nominal current is also imposed at the extraction terminal. The current in the single strands is not therefore imposed and is only determined by the inductive and resistive terms of the DAE system.

Another thing to note is the fact that when the cable operates in nominal conditions, the entirety of the current flows through the superconducting material. This means that there is no transverse current transfer between different stacks and the transverse conductances matrix is null:

$$\mathbf{G} = 0 \quad (5.6)$$

In our model, the DAE system covers only the time ramp from 0 to 50 kA and not the subsequent flat-top. The ramp takes 250 s to complete and is analyzed with a time step $\Delta T = 10 \text{ s}$, giving $N_t = 25$ steps. The DAE is advanced using an implicit first-order scheme, with each time step solving:

$$\left(\frac{\mathbf{M}}{\Delta t} + \mathbf{K} \right) \mathbf{x}^{n+1} = \frac{\mathbf{M}}{\Delta t} \mathbf{x}^n + \mathbf{n} + \mathbf{1} \quad (5.7)$$

Because of the dependence of the resistive term on current, the implicit problem is non-linear. To solve it, the code implements a Picard/fixed-point iteration. At each step the current estimate of $\mathbf{i}$ is used to recalculate the resistance. The linearized DAE is then solved and the currents are updated. This is repeated until convergence, with the convergence condition set as:

$$\frac{|\mathbf{x}^{k+1} - \mathbf{x}^k|}{|\mathbf{x}^{k+1}|} < 10^{-8} \quad (5.8)$$

It's important to not that the DAE is not an electro-thermal model. All the quantities are evaluated at constant $T = T_{op}$. The thermal evaluation is instead presented in chapter 8.

5.3 Equipartition verification

Because of the geometry of the cable, the inductive and resistive terms should be approximately the same for each strand. An ideal equipartition would give, at the end of the ramp:

$$I_{ideal} = \frac{50000}{6} = 8333.3 \text{ A} \quad (5.9)$$

The actual computed strand currents exhibit a maximum deviation from this ideal behavior of only 0.651%. The code checks that the total current is conserved:

$$\sum_{j=1}^6 I_j = I_{nom} \quad (5.10)$$

The resulting error is 4.37×10^{-16} . The free-row residual, which verifies that the solved DAE equations are numerically satisfied on the unconstrained rows, is 1.51×10^{-14} . Both constraints are therefore satisfied essentially to machine precision. Since there are no transverse currents, the code also checks that the current is conserved along each strand.

Lastly, the code also features a check to show that the equipartition arises only from the geometry and not from other factors like the imposed boundary conditions. It does so by solving a system with deliberately perturbed critical-current profiles, making the strands not electrically equivalent. The result shows that the resulting currents become strongly nonuniform between different strands, demonstrating the importance of the cable geometry for equipartition.

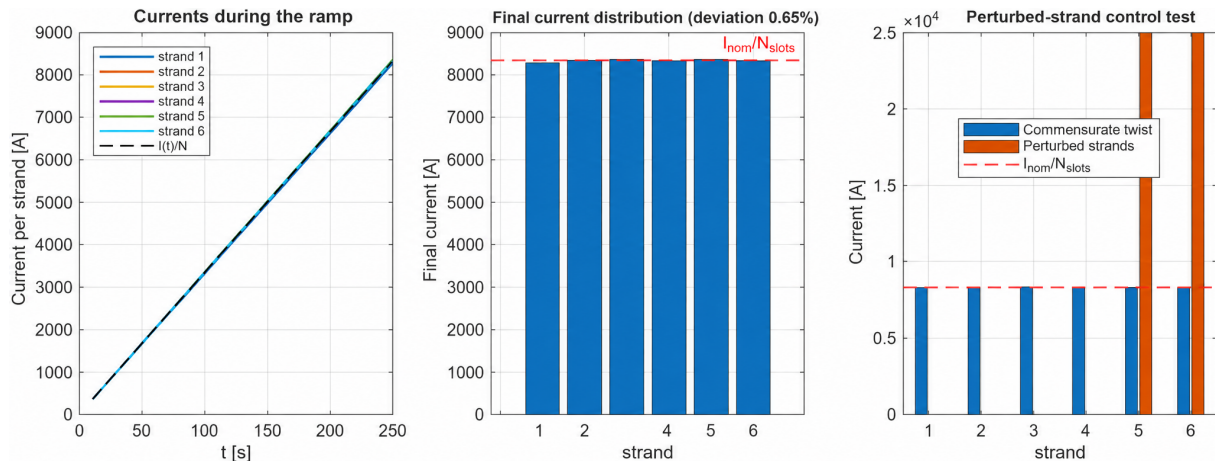


Figure 5.1: Verification of current equipartition among the six equivalent tape stacks. (a) Evolution of the strand currents during the 0–50 kA ramp, compared with the ideal distribution $I(t)/N_{\text{slots}}$. (b) Final current distribution at 50 kA, showing a maximum deviation of 0.65% from exact equipartition. (c) Control case obtained by deliberately perturbing the strand critical-current properties, which produces a strongly nonuniform current distribution and demonstrates that equipartition in the nominal case is a consequence of strand equivalence rather than an imposed numerical constraint.

Table 5.1: DAE numerical settings and equipartition result.

Quantity	Value
Ramp duration	250 s
Time step	10 s
Number of time steps	25
Mean Picard iterations per step	5.0
Estimated condition number	2.15×10^9
Ideal strand current	8333.3 A
Final strand currents	8279.1, 8334.9, 8359.7, 8331.0, 8360.6, 8334.8 A
Maximum equipartition deviation	0.6509%
Relative total-current error	4.37×10^{-16}
Free-row relative residual	1.51×10^{-14}
Perturbed-case deviation	200%

Chapter 6

CICC Cross-Section Layout

6.1 Cross-section design method and assumed construction parameters

As previously mentioned, the six equivalent strands that we used for the electromagnetic model correspond physically to 80 stacked tapes. Each of these stacks is assumed to completely fill the slot they are in, meaning they are in perfect contact with the copper former surrounding them. This means there is no helium region surrounding the tape: the only fluid region is the central circular channel. The copper former is encased in a stainless steel jacket, whose purpose is maintaining the structural integrity of the cable. We assumed the thickness of the jacket to be $t_{jacket} = 2 \text{ mm}$. The width of the tapes is assumed to be $w_{slot} = 4 \text{ mm}$, while the thickness of a single tape is $t_{tape} = 55.6 \text{ }\mu\text{m}$. The thickness of an entire stack is therefore $d = 80t_{tape} = 4.448 \text{ mm}$. The thickness of the copper wall between the tape stacks and the steel jacket $t_{web,out}$ as well as the thickness of the copper wall between the tape stacks and the helium channel $t_{web,in}$ were both assumed to measure 1 mm .

6.2 Derived slot geometry and geometric consistency

Starting from the outside radius of the cable $r_{cab} = 12.50 \text{ mm}$, we can obtain the radius of the copper former by subtracting the steel jacket thickness:

$$r_{former,out} = r_{cab} - t_{jacket} = 10.5 \text{ mm} \quad (6.1)$$

Other significant radii that have been used in calculations are the outer slot radius:

$$r_{slot,out} = r_{former,out} - t_{web,out} = 9.5 \text{ mm} \quad (6.2)$$

The radius at the slot centroids:

$$R_{core} = r_{slot,out} - \frac{d_{slot}}{2} = 7.276 \text{ mm} \quad (6.3)$$

And the inner slot radius:

$$R_{core} - \frac{d_{slot}}{2} = 5.052 \text{ mm} \quad (6.4)$$

We can also define the equivalent strand radius as:

$$r_0 = \sqrt{\frac{A_{stack}}{\pi}} = 2.38 \text{ mm} \quad (6.5)$$

Lastly, the central channel radius which will be used to derive other quantities in the thermo-hydraulic model, is:

$$r_{channel} = r_{near} - t_{web,in} \quad (6.6)$$

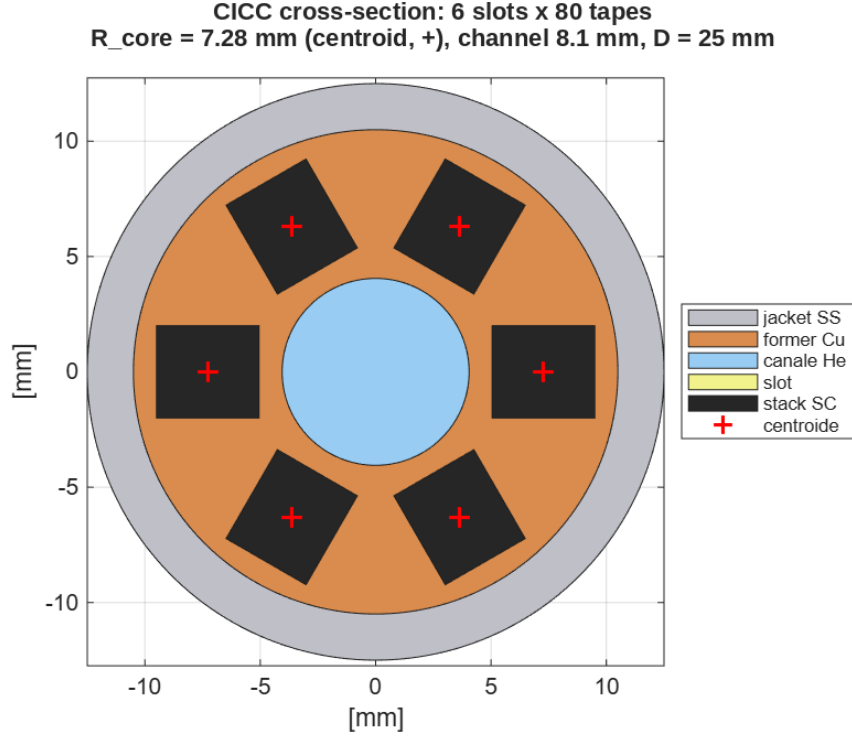


Figure 6.1: Cross-section of the adopted HTS CICC. The conductor consists of six REBCO tape stacks embedded in a copper former, surrounded by a stainless-steel jacket and cooled by a central helium channel. Each stack contains 80 tapes and is represented by its centroid in the conductor-level electromagnetic model.

Table 6.1: Final CICC cross-section dimensions.

Quantity	Value
Cable radius	12.50 mm
Stainless-steel jacket thickness	2.00 mm
Outer copper web	1.00 mm
Inner copper web	1.00 mm
Slot depth / stack depth	4.448 mm
Slot width	4.00 mm
R_{core}	7.276 mm
Equivalent strand radius r_0	2.380 mm
Inner stack edge r_{near}	5.052 mm
Central He-channel diameter	8.104 mm

Chapter 7

AC Losses During the Current Ramp

7.1 Loss models and validity checks

In our model, AC losses represent the dissipation caused by the variation in time of the current and the magnetic field during the ramp from 0 to 50 kA. These losses are not evaluated by the DAE model and require specific analytical formulas. This is because the DAE model doesn't account for transverse currents and lumps the tape stacks as single strands. This means that it cannot describe the magnetic flux penetration across the width of a single REBCO tape and is not therefore able to evaluate hysteresis or coupling losses. We approximate the magnetic field to scale linearly with the current during the ramp:

$$B(s, t) = \lambda(t)B_f(s) \quad (7.1)$$

Where B_f is the field reached at the end of the ramp and $\lambda(t) = I(t)/I_{nom}$. Since the ramp is linear, we can write:

$$\frac{\partial B}{\partial t} = \frac{B_f(s)}{t_{ramp}} \quad (7.2)$$

The hysteresis and coupling contributions make up the AC losses heat source used in the thermal model:

$$q_{AC} = q_{hyst} + q_{coup} \quad (7.3)$$

For the hysteresis contribution, the model uses the following equation:

$$P_{hyst,tape} = \frac{2I_c d_{eff}}{3\pi} \left| \frac{dB_{\perp}}{dt} \right| \quad (7.4)$$

This expression is meant for superconducting tapes in the fully penetrated critical-state regime. We can assume that full penetration is reached since the peak field is much larger than the penetration field B_c :

$$\frac{B}{B_c} = \frac{B}{\frac{\mu_0 I_c}{\pi w}} \approx 985 \quad (7.5)$$

For a wide thin tape subjected to perpendicular field we can assume $d_{eff} = w_{tape} = 4 \text{ mm}$. Only the perpendicular component of the field is relevant for hysteresis losses as this is the component producing flux penetration across the face of the tape. The code explicitly computes this component using the local tape orientation. Twist geometry therefore directly influences AC losses. This component is not, however, influenced by the power law coefficient $n(T, B)$. This model is based on the Bean critical-state approximation,

which does not require the finite exponent used in the DAE model. This also means that the error associated with n values at fields higher than 8 T doesn't propagate into this calculation. The total hysteresis contribution of a stack is evaluated assuming that all 80 tapes contribute independently:

$$P_{hyst,stack} = 80P_{hyst,tape} \quad (7.6)$$

This is a conservative assumption, as in reality the mutual magnetic shielding between tapes would reduce the total magnetization loss. The maximum resulting magnetization per equivalent strand is:

$$P_{hyst,max} = 0.5619 \text{ W/m} \quad (7.7)$$

The inter-strand coupling losses are caused by induced coupling currents which flow transversely through the metal. These closed currents are induced by dB/dt and follow transverse conductive paths even when the supplied current remains entirely in the superconductor. The adopted model for these losses is:

$$\frac{n_g \tau}{\mu_0} \left(\frac{dB}{dt} \right)^2 A_{stack} \quad (7.8)$$

Where:

$$\tau = \frac{\mu_0 (l_p/2\pi)^2}{2\rho_{eff}} \quad (7.9)$$

n_g is the geometric shape factor, which for a transverse field is set as $n_g = 2$. A_{stack} is the cross-sectional area associated with a single tape stack, while l_p is the twist pitch. Since l_p increases as the winding radius increases, coupling losses also become more sensitive at larger radii. ρ_{eff} is defined by the Kohler's rule:

$$\rho_{eff}(B) = \rho_{Cu,0} MR(B) g_{geom} \quad (7.10)$$

Where $\rho_{Cu,0} = 1.55 \times 10^{-10} \text{ } \Omega m$ is the low-temperature zero-field resistivity for $RRR = 100$, $MR(B) = 1 + 10^{-2.69} (|B|RRR)^{1.055}$ is a term that accounts for the magnetoresistance of copper and g_{geom} is a term accounting for the actual transverse current path through the copper former. This last term is needed because the transverse current has to flow around the stack slots and the central helium channel, meaning its path is longer and has less conductive area than in a former with no interruptions. To obtain an effective transverse conductivity, the code solves a two-dimensional conductivity problem:

$$\nabla \cdot (\sigma \nabla \phi) = 0 \quad (7.11)$$

As was previously stated, the model assumes perfect contact between the tape stacks and the copper former. This, however, causes the coupling losses to increase, as the transverse resistance is lower and:

$$P_{coup} \propto \frac{1}{\rho_{eff}} \quad (7.12)$$

An important assumption on which the model is based is that the induced current loops respond on a much shorter time scale compared to the current ramp time. Since τ is the time constant of the problem, the model checks:

$$\frac{\tau_{max}}{t_{ramp}} = \frac{17.16}{250} = 0.0686 < 0.1 \quad (7.13)$$

Therefore the quasi-stationary approximation is acceptable. The model also evaluates transport-current hysteresis losses and eddy-current losses, but both turn out to be negligible compared to the two other contributions.

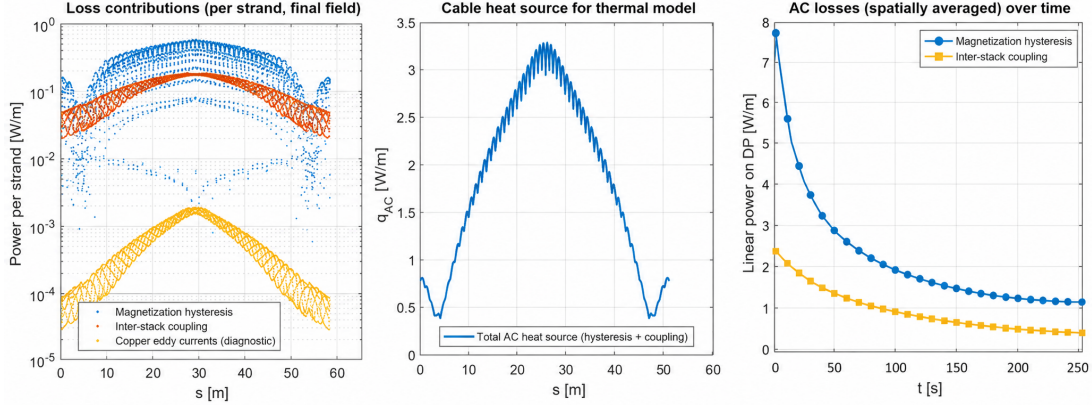


Figure 7.1: AC-loss distribution in the central double pancake during the 0–50 kA current ramp. Left: spatial distribution of the principal loss contributions per equivalent strand at final field. Copper eddy-current losses are shown as a diagnostic but are negligible compared with the magnetization and coupling terms. Center: total cable heat source $q_{AC}(s)$, obtained by summing magnetization and coupling losses over the six tape stacks and subsequently supplied to the thermo-hydraulic model. Right: time evolution of the spatially averaged magnetization and coupling losses during the 250 s linear current ramp.

7.2 Time dependence and quench-protection trade-off

Although a linear current ramp means that the time derivatives of current and field are constant, it does not mean that AC power is constant during the ramp.

For the hysteresis losses we have:

$$P_{hyst} \propto I_c(T_{op}, B) \dot{B}_\perp \quad (7.14)$$

Since the field changes during the ramp, the critical current I_c also does, decreasing as the magnetic field increases. This means that the hysteresis losses are maximum at the beginning of the ramp and then progressively decrease.

For the coupling losses we have:

$$P_{coup} \propto \tau(B) \dot{B}^2 \quad (7.15)$$

With:

$$\tau(B) \propto \frac{1}{\rho_{eff}(B)} \quad (7.16)$$

Since the copper magnetoresistance increases with the magnetic field, the coupling losses also decrease throughout the ramp. Since both contributions decrease during the ramp, the final AC power is lower than the average one on the whole ramp. The average AC power is:

$$\bar{P}_{AC} = 179.5 \text{ W} \quad (7.17)$$

And therefore the deposited energy is:

$$E_{AC} = \bar{P}_{AC} t_{ramp} \approx 44.9 \text{ kJ} \quad (7.18)$$

While the final AC power comes to:

$$P_{AC,f} = 94.7 \text{ W} \quad (7.19)$$

Table 7.1: Main AC-loss results for the completed reference run.

Quantity	Value
Peak-field tape I_c	153.0 A
Full-penetration ratio B/B_c	985
Maximum hysteresis loss per strand	0.5619 W/m
Maximum coupling loss per strand	0.1846 W/m
Maximum transport loss per strand	1.49×10^{-4} W/m
Maximum Cu eddy loss per strand	1.97×10^{-3} W/m
Maximum cable heat source	3.290 W/m
Final-field hysteresis power, whole DP	62.8 W
Final-field coupling power, whole DP	31.8 W
Final-field total AC power, whole DP	94.7 W
Ramp-average AC power, whole DP	179.5 W
Ramp-integrated AC energy	44.9 kJ
Maximum coupling time constant over ramp	17.16 s
$\tau_{\max}/t_{\text{ramp}}$	0.0686

Chapter 8

Thermo-Hydraulic Model and Thermal-Margin Verification

8.1 GANDALF model

The implemented thermo-hydraulic model is a GANDALF simulation. Although this type of model is made for LTS cables, since the CICC layout we are using has a one region cooling channel, we can approximate the thermo-hydraulics of our architecture using this model. At every axial position along the conductor, four temperature values are retained: one for the jacket T_j , one for the copper former T_{Cu} , one for the superconducting tape stacks T_{sc} and one for the bulk helium T_{He} . The cross section is represented by these four thermal regions, whose temperatures changes spatially and in time. The four temperatures are independently solved by the code.

While the thermal variables undergo a transient evolution, pressure and velocity are treated quasi-statically. This assumption is based on the acoustic time, which is the characteristic time needed for a pressure signal to travel through the helium channel. We define it as:

$$t_{ac} = \frac{L_{branch}}{c} \quad (8.1)$$

Where L_{branch} is the length of one hydraulic branch and c is the local speed of sound in helium. In this case we find $t_{ac} \approx 0.093 \text{ s}$, which is a timescale several orders of magnitude smaller than the duration of the current ramp. This means that pressure disturbances equilibrate many orders of magnitude faster than the thermal evolution and is the justification for not solving a fully transient system. However, the transit time of helium through the branch in the minimum-flow point is $t_{transit} \approx 98 \text{ s}$, meaning that fluid transport cannot be considered instantaneous. This is represented by the axial advection term in the helium energy equation.

The model uses temperature-dependent properties for the solid materials and real thermodynamic data for the helium, although they are evaluated at the reference outlet pressure and not at the locally reconstructed pressure.

8.2 Governing heat-transfer terms

The four governing equations of the model are:

$$\rho_j c_{p,j} A_j \frac{\partial T_j}{\partial t} = \frac{\partial}{\partial s} \left(k_j A_j \frac{\partial T_j}{\partial s} \right) + G_{j-Cu} (T_{Cu} - T_j), \quad (8.2)$$

$$\begin{aligned} \rho_{Cu} c_{p,Cu} A_{Cu} \frac{\partial T_{Cu}}{\partial t} = & \frac{\partial}{\partial s} \left(k_{Cu} A_{Cu} \frac{\partial T_{Cu}}{\partial s} \right) + G_{sc-Cu} (T_{sc} - T_{Cu}) + G_{j-Cu} (T_j - T_{Cu}) \\ & + h_{He} P_{wet} (T_{He} - T_{Cu}) + q'_{coup}(s, t), \end{aligned} \quad (8.3)$$

$$\rho_{sc} c_{p,sc} A_{sc} \frac{\partial T_{sc}}{\partial t} = \frac{\partial}{\partial s} \left(k_{sc} A_{sc} \frac{\partial T_{sc}}{\partial s} \right) + G_{sc-Cu} (T_{Cu} - T_{sc}) + q'_{hyst}(s, t), \quad (8.4)$$

$$\rho_{He} c_{p,He} A_{He} \frac{\partial T_{He}}{\partial t} + \dot{m}_b c_{p,He} \frac{\partial T_{He}}{\partial s} = h_{He} P_{wet} (T_{Cu} - T_{He}). \quad (8.5)$$

We can observe that the two AC loss heat terms from the previous chapter are present. The hysteresis contribution originates in the tapes so is deposited in the superconductor equation, while coupling losses are induced by currents circulating through the former and are deposited in the copper equation. Both of these terms disappear after $t > 250$ s. The conductances between superconductor, copper and jacket are evaluated as:

$$G_{sc-Cu} = \frac{P_{sc}}{l_{sc}/k_{sc} + l_{Cu,in}/k_{Cu}} \quad (8.6)$$

$$G_{Cu-j} = \frac{P_{Cu-ss}}{l_{Cu,out}/k_{Cu} + l_j/k_j} \quad (8.7)$$

Since the materials are in perfect contact there is no additional contact-resistance term present.

The convective conductance per unit length of helium is $H = h_{He} P_{wet}$ with:

$$h_{He} = \frac{Nu k_{He}}{D_h} \quad (8.8)$$

The model evaluates Nu and Pr numbers locally and uses a smooth-tube correlation. For laminar flow $Nu = 4.36$, while for turbulent flow the Petukhov/Gnielinsky correlation is used, interpolating in the transition range. At the final base operating point we have $Re \approx 7 \times 10^4$, so the flow is turbulent.

8.3 Flow arrangement, boundary conditions and numerical discretization

The DP has a single helium inlet located at the inner interconnection between the two pancakes. The flow then splits evenly in the two branches, so $\dot{m}_A = \dot{m}_B = \frac{\dot{m}_{tot}}{2}$. The helium temperature is imposed at the inlet of the two branches as $T_{in} = 4.5$ K, while no outlet temperature is prescribed since the advection equation determines it. For the solid regions, the model imposes zero axial heat flux at both ends of each branch. Pressure is imposed at the outlet as $p_{out} = 10$ bar and is then reconstructed upstream.

The three solid domains are discretized using centered finite differences for axial conduction. At the interfaces, the kA terms are averaged using harmonic averaging. Helium advection uses instead a first-order backward difference since the flow is directed from the inlet to the outlet.

The solver uses a semi-implicit Backward Euler method for time integration, with the temperature-dependent material properties being evaluated at the previous time step and held fixed during the new linear solve. The time step used in the final convergence run is $\Delta t = 0.625$ s. The thermal simulation continuous for 100 s after the ramp is concluded, in order to account for thermal inertia.

Table 8.1: Thermo-hydraulic geometry and numerical settings.

Quantity	Value / method
Central DP length	51.60 m
Hydraulic branch length	25.80 m
He flow area	51.58 mm ²
Hydraulic diameter	8.104 mm
Wetted perimeter	25.46 mm
Baseline inlet temperature	4.50 K
Reference outlet pressure	10.0 bar
Maximum total mass flow	15.0 g/s
Acoustic time per branch	0.0930 s
Ramp duration	250 s
Simulation end time	350 s
Thermal time step in convergence run	0.625 s
Latest dense convergence target range	$\Delta x \approx 0.005$ m to 1.6 m
Solid spatial scheme	centered finite differences
Helium advection scheme	first-order backward/upwind
Time integration	semi-implicit Backward Euler

8.4 Minimum mass-flow search and final thermal solution

The governing headline requirement is;

$$\Delta T_{th,min} = \min[T_{cs}(s) - T_{sc}(s, t)] \geq 5 \text{ K} \quad (8.9)$$

Since T_{cs} does not vary with time in the model, the code first finds the maximum value in time for T_{sc} and then evaluates the temperature difference:

$$\Delta T(s) = T_{cs}(s) - T_{sc,max}(s) \quad (8.10)$$

The minimum value in this profile is the reported thermal margin. The mass flow is then varied until the smallest flow satisfying the thermal and pressure constraints is found. This is done using bisection with a tolerance of 0.001 g/s. With an inlet temperature of 4.5 K, this results in approximately 4 g/s of helium mass flow, which accounts for only $\approx 27\%$ of the mass flow budget. The main results are reported in Table 8.2. The most critical position is found at $s = 11.83$ m. The solid materials all reach a maximum

temperature of $\approx 16 T$, while the pressure drop across the branch is very low at 0.0094 bar , given an inlet pressure of $\approx 10 \text{ bar}$.

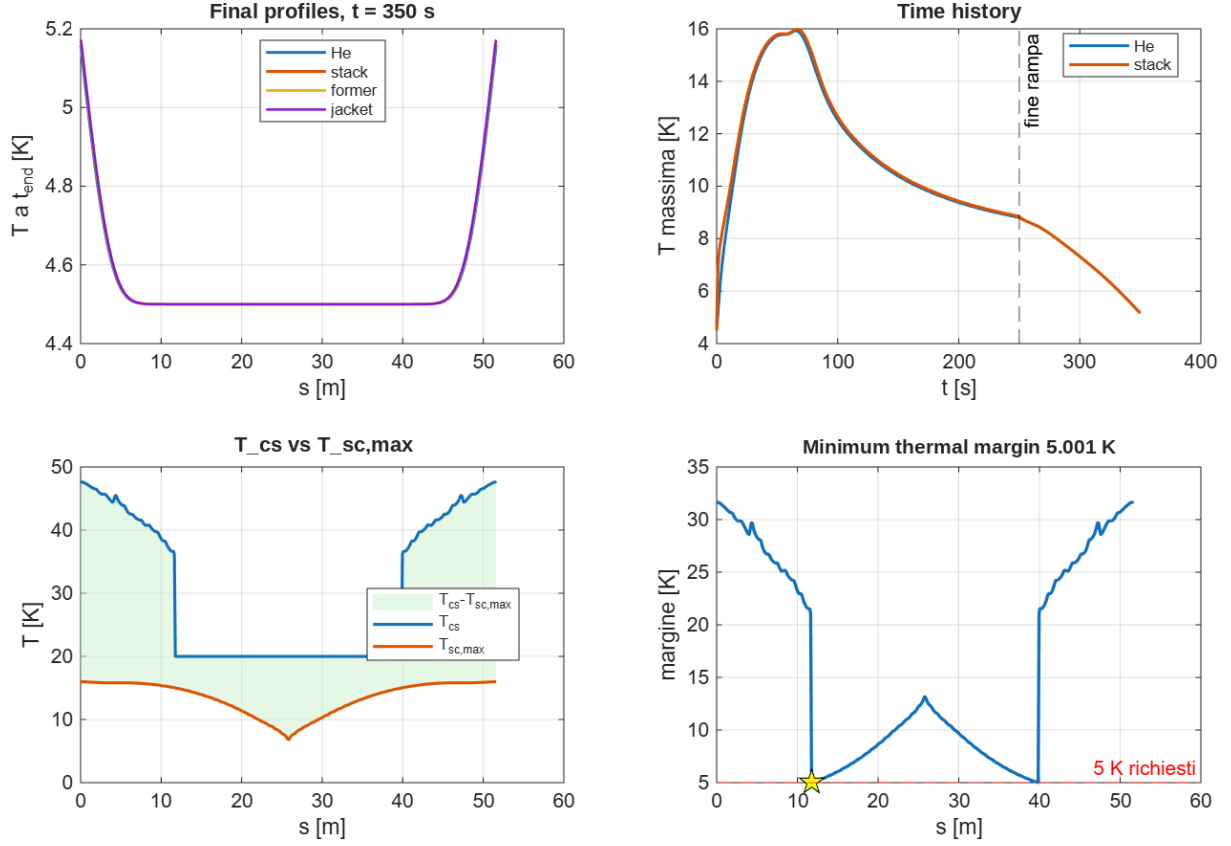


Figure 8.1: Cleaned four-panel thermo-hydraulic summary: final temperature profiles, temperature history, T_{cs} versus maximum stack temperature, and transient thermal margin.

Table 8.2: Main thermo-hydraulic results from the last completed stable reference run.

Quantity	Value
Minimum total He mass flow	4.001 g/s
Minimum transient thermal margin	5.004 K
Location of minimum margin	11.83 m
Maximum stack temperature	16.005 K
Maximum copper-former temperature	16.004 K
Maximum helium temperature	15.934 K
Branch pressure drop	0.0094 bar
Maximum inlet pressure at base point	10.009 bar
Base-point Reynolds-number range	6.78×10^4 – 7.44×10^4
Base-point He velocity range	0.262 m/s to 0.277 m/s
Base-point transit time	97.9 s
$T_{in,max}$ at 15 g/s	11.175 K
Frontier-point inlet pressure at 15 g/s	10.126 bar

8.5 Spatial convergence

The thermo-hydraulic results are verified using a spatial-convergence study. The time step is kept fixed while the spatial step is progressively reduced. The resulting log-log trend shows a convergence rate approximately of the first order. Although, the solids use centered discretization, which is of the second order, the helium equation uses first-order upwind advection. Since the four equations are coupled, the first-order helium discretization dominates the global error.

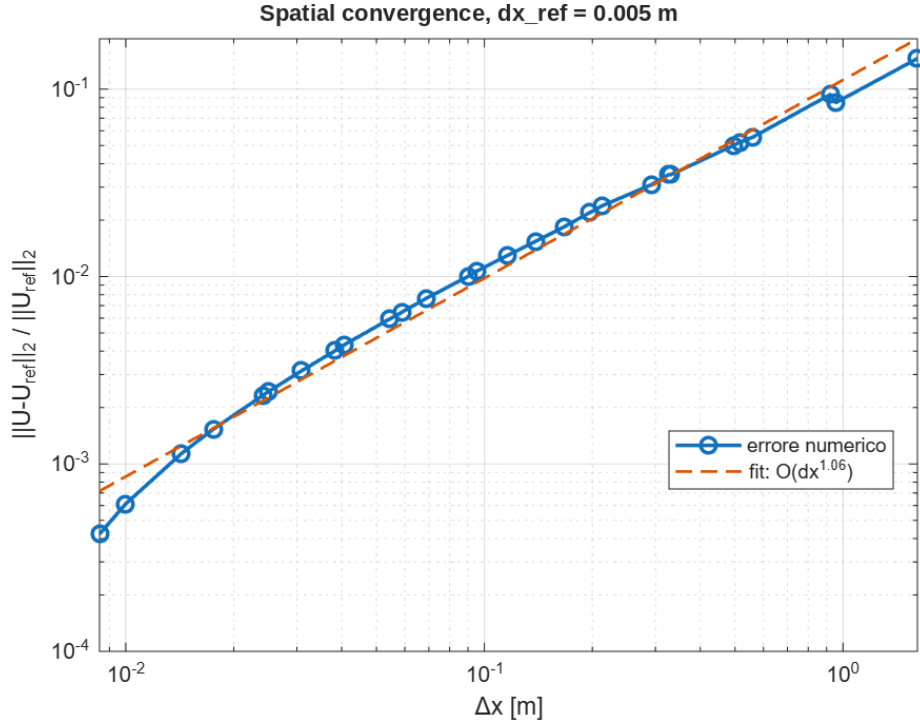


Figure 8.2: Completed spatial-convergence study from the reference run, showing an observed order of approximately 1.10. The latest main uses the same error definition with a denser Δx set.

8.6 Operating frontier

Since the project headline provides a range for the inlet temperature of helium and not a specific value, it is possible to design a feasibility frontier at different values for T_{in} . We can observe that, as the mass flow increases, the heat-removal capability of helium also increases, allowing for higher inlet temperature while still satisfying the thermal margin. When the maximum available flow of 15 g/s is used, the inlet temperature reaches 11.175 K. The various operating points are reported in Figure 8.3 and Table 8.3.

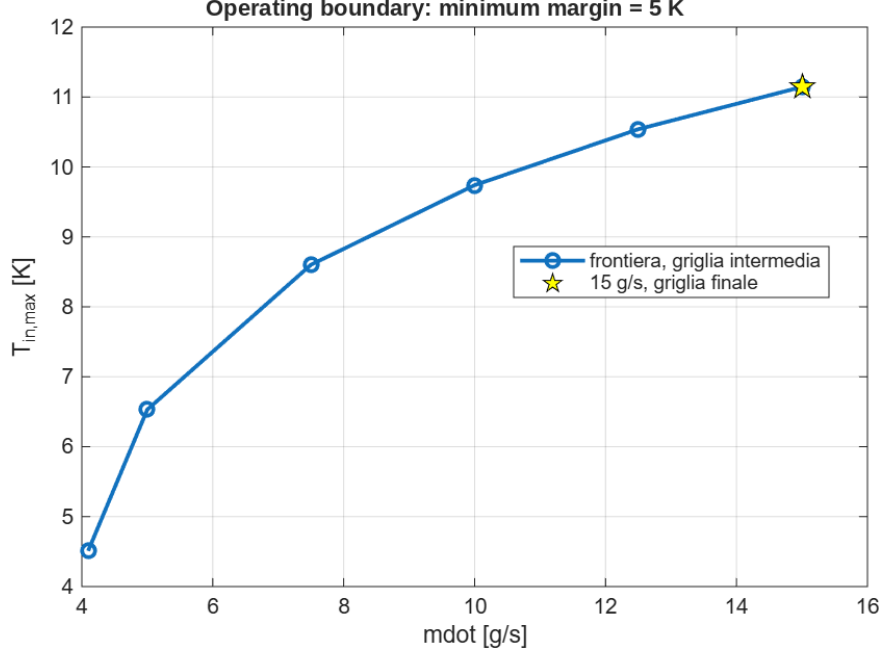


Figure 8.3: Operating frontier between total helium mass flow rate and maximum allowable inlet temperature for a 5 K minimum transient thermal margin.

Table 8.3: Operating-frontier points from the completed reference run.

Total mass flow [g/s]	$T_{in,max}$ [K]	Inlet pressure [bar]
4.001	4.500	10.009
5.000	7.002	10.015
7.500	8.878	10.034
10.000	9.986	10.059
12.500	10.765	10.090
15.000	11.329	10.127
15.000, fine-grid check	11.175	10.126

8.7 Temperature distribution along the pancake

The spatial distribution of the temperature provides a direct interpretation of the results of the thermo-hydraulic model. Since helium is supplied at the central interconnection and the total mass flow is divided between the two halves of the double pancake, each branch is cooled from the inner connection toward its outer end. The helium temperature therefore increases progressively along the flow direction, as heat is transferred from the conductor to the coolant.

The superconducting stack and the copper former remain at very similar temperatures throughout the transient. This is a consequence of the strong thermal coupling assumed between the two regions and is also reflected by their almost identical maximum temperatures, $T_{sc,max} = 16.005$ K and $T_{Cu,max} = 16.004$ K. The helium reaches a slightly lower maximum temperature of approximately 15.934 K, as expected from the finite convective temperature difference required to remove heat from the conductor.

The thermal margin is not determined solely by the position of maximum temperature. At each axial location it is defined as

$$\Delta T(s) = T_{cs}(s) - T_{sc,max t}(s),$$

so that both the local thermal response and the spatial variation of the current-sharing temperature contribute to the limiting condition. The minimum value,

$$\Delta T_{\min} = 5.004 \text{ K},$$

occurs at approximately $s = 11.83 \text{ m}$. The limiting region is therefore the result of the combined electromagnetic and thermal distributions rather than simply the hottest point of the conductor.

The temperature maps also show that the center-fed configuration produces a nearly symmetric thermal response in the two hydraulic branches. After the end of the 250 s current ramp, the AC-loss source vanishes, while the conductor and helium temperatures continue to evolve because of the thermal energy stored in the solids and coolant. The additional post-ramp simulation time is therefore required to ensure that the actual maximum temperature and minimum current-sharing margin are captured.

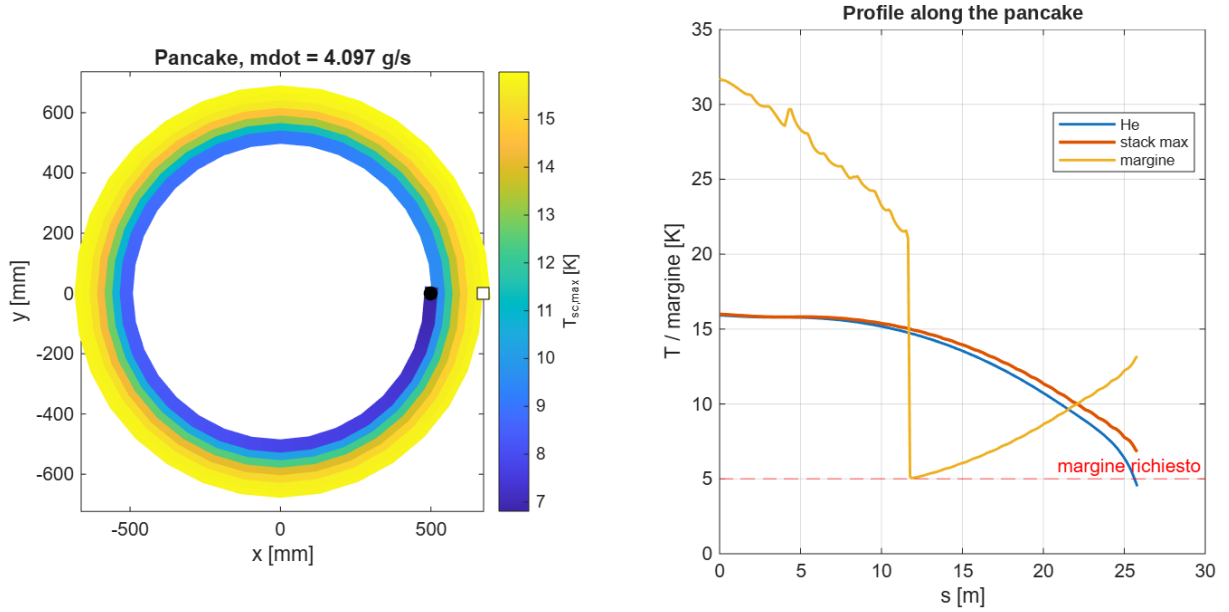


Figure 8.4: Maximum stack-temperature distribution along the pancake and the corresponding helium, stack, and thermal-margin profiles along one hydraulic branch.

We can observe that the GANDALF model is 1D and describes axial heat transfer only, by the fact that on a cable section temperature doesn't significantly change.

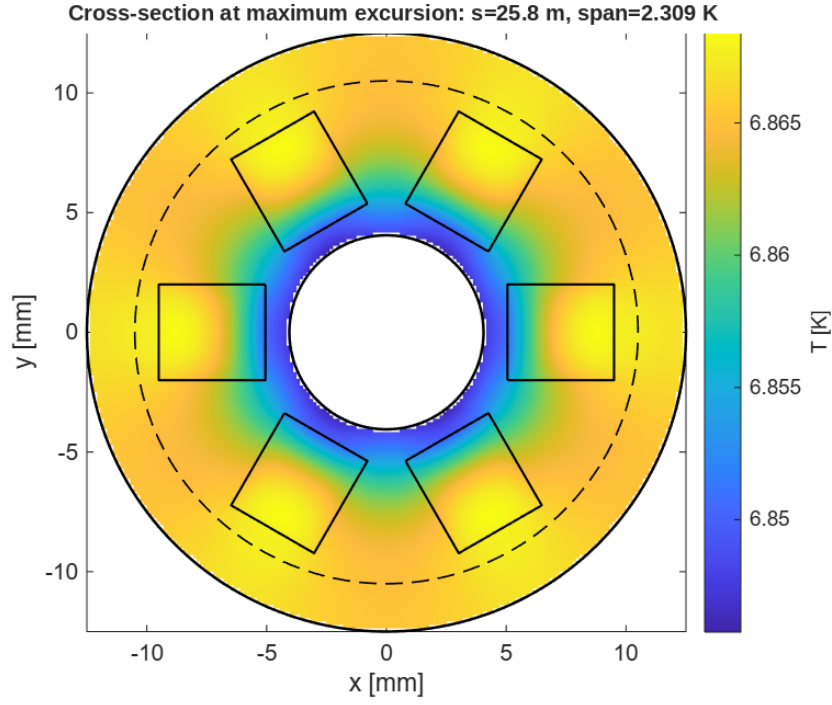


Figure 8.5: Diagnostic reconstruction of the temperature distribution across the CICC cross-section at the location of maximum transverse thermal excursion. The temperature variation across the section remains small compared with the axial temperature evolution of the conductor, supporting the use of a one-dimensional thermo-hydraulic model in which the dominant temperature gradients are resolved along the conductor coordinate s . The map is used only as a cross-sectional consistency check and is not an additional two-dimensional GANDALF solution.

Chapter 9

Conclusions and model Limitations

The coupled electromagnetic and thermo-hydraulic analysis led to a feasible reference design consisting of 29 double pancakes with seven radial turns per pancake, operated at a nominal current of 50 kA. The resulting peak magnetic field on the conductor is approximately 15.07 T, in agreement with the target value of about 15 T. The conductor is based on six equivalent REBCO tape stacks, each containing 80 tapes, and the non-linear electrical model confirms that the nominal current is distributed almost uniformly among the six stacks, with a maximum deviation from ideal equipartition of only 0.65%. During the 250 s current ramp, magnetization hysteresis and inter-stack coupling were identified as the relevant AC-loss mechanisms, resulting in a final-field power of 94.7 W, a ramp-averaged power of 179.5 W, and a total deposited energy of approximately 44.9 kJ in the central double pancake. The resulting heat load was then applied to the transient thermo-hydraulic model. For a helium inlet temperature of 4.5 K, the minimum total mass flow required to satisfy the design condition $\min_{s,t}[T_{cs}(s) - T_{sc}(s,t)] \geq 5$ K is approximately 4.00 g/s, well below the available 15 g/s limit. The corresponding minimum transient margin is 5.004 K, while the maximum stack temperature remains close to 16.0 K and the calculated hydraulic pressure drop is negligible compared with the allowed pressure range. Moreover, at the maximum available flow rate, an inlet temperature of approximately 11.18 K can still be tolerated while preserving the required margin. These results show that, within the assumptions of the adopted electromagnetic, AC-loss and reduced thermo-hydraulic models and for the nominal superconducting-property dataset, the selected configuration simultaneously satisfies the principal magnetic-field and thermal-management requirements of the project.

The conclusions above remain subject to the assumptions and limitations of the adopted models. The critical-current description is based on the supplier lift-factor dataset and on the conservative reference value $I_{c,77\text{K},sf} = 100$ A, while the detailed $n(T, B)$ measurements extend only to 8 T; above this field, the electrical model retains the highest available n -value rather than extrapolating it. The laboratory measurements were therefore used primarily to validate the experimental I - V analysis procedure and qualitative field dependence, rather than as quantitative design data. The final transient 5 K thermal margin is obtained using the nominal T_{cs} profile; the -20% lift-factor case introduced as a robustness sensitivity was not propagated through the complete thermo-hydraulic transient. Furthermore, the electrical model represents the conductor through six equivalent tape stacks rather than individual tapes, the magnetization-loss calculation adopts a Bean critical-state approximation, and the thermo-hydraulic analysis uses a reduced one-dimensional slow-transient formulation with ideal stack-former thermal contact and

a nominal 50/50 helium-flow split. Detailed quench propagation, current transfer following a local transition, structural stresses and manufacturing-dependent contact resistances are outside the present scope. Future work should therefore focus on high-field characterization of $n(T, B)$, experimental determination of the electrical and thermal stack-former contact properties, validation of the proposed hydraulic configuration, and, if required, extension to a fully coupled electro-thermal quench model.

Within these stated assumptions and within the electromagnetic and thermo-hydraulic scope of the project, the proposed configuration provides a consistent and feasible solution for achieving the required high magnetic field while maintaining the prescribed thermal margin during the nominal current ramp.

Bibliography